ByungUk Park

bpark74@wisc.edu | profile | publications | byungukP | byungukP

RESEARCH INTERESTS

I develop physically grounded machine learning methods for biomolecular systems, with interests spanning geometric deep learning, generative modeling, and molecular dynamics. My work focuses on characterizing protein conformational landscapes and designing physically feasible proteins under structural or functional constraints, grounded in rigorous simulation-based validation.

EDUCATION

University of Wisconsin–Madison, Madison, WI — Sep 2021–Present
Ph.D. Biophysics (Advisor: Reid C. Van Lehn)
M.S. Biophysics, 2023

Seoul National University, Seoul, South Korea — Mar 2014–Feb 2021
B.S. Biological Sciences (*Cum Laude*)
Military service leave (Aug 2016 – Apr 2018)

WORK EXPERIENCE

Van Lehn Group – UW–Madison — Madison, WI, USA
Graduate Student Researcher — Sep 2021 — Present

- Developed physically grounded machine learning models combining geometric deep learning and molecular dynamics to predict and design biomolecular interfaces.
- Built scalable pipelines integrating simulation and machine learning to extract binding modes, conformational dynamics, energetics, and specificity of complex biomolecular systems.
- Applied enhanced sampling techniques (umbrella sampling, replica exchange, metadynamics) to investigate interaction energetics of biomolecules (*e.g.*, protein, membrane, drug molecule).
- Conducted large-scale GPU simulations and model training using PyTorch and CUDA-accelerated HPC workflows.

Seoul National University — Seoul, South Korea
Undergraduate Research Assistant

- Laboratory of Biophysical Chemistry (PI: JunSeock Koh) — Sep 2019 — Dec 2020
  : Performed isothermal titration calorimetry (ITC) and X-ray crystallography to characterize binding patterns of intrinsically disordered proteins (IDPs)
- Laboratory of Molecular Regenerative Medicine (PI: Jin-Hong Kim) — Feb 2019 — Aug 2019
- Laboratory of Microbial Physiology (PI: Yeong-Jae Seok) — Dec 2015 — Jan 2016

PUBLICATIONS

1. **B. Park**, R. C. Van Lehn. *Decoding Protein–Membrane Binding Interfaces from Surface-Fingerprint-Based Geometric Deep Learning and Molecular Dynamics Simulations.* **J. Chem. Inf. Model.**, 2026, 66, (4), 2299–2310. [Link]
2. **B. Park**, R. C. Van Lehn. *Surface-Aware Generative Design of Selective Binders for Protein–Membrane Interface.* 2026, in preparation.
3. A. Mukherjee, **B. Park**, A. Malmstrom, R. C. Van Lehn, V. M. Zavala. *Scalable Extraction of Information on Protein-Protein Interactions using Topological Data Analysis.* **J. Chem. Inf. Model.**, 2026, manuscript submitted. [Preprint]
4. C. Qiu, **B. Park**, F. Mukherjee, N. A. Biok, R. C. Van Lehn, S. H. Gellman, N. L. Abbott. *Influence of Charged Residue Identity on Hydrophobic Interactions mediated by Amphiphilic Helices of α-Peptides.* 2026, in preparation.
5. S. Walters, **B. Park**, C. Labrecque, F. Musayev, R. C. Van Lehn, B. Fuglestad. *Revealing interactions between glutathione peroxidase 4 and phosphoinositides.* **Protein Sci.**, 2026, manuscript submitted. [Preprint]
6. **B. Park**, R. C. Van Lehn. *Bioinspired Design Rules for Flipping across the Lipid Bilayer from Systematic Simulations of Membrane Protein Segments.* **Mol. Sys. Des. Eng.**, 2025, 10, 567-584. [Link]
7. J. Morstein, **B. Park**, A. Capecchi, K. Hinnah, J. Petit-Jacques, R. C. Van Lehn, J-L Reymond, D. Trauner. *Medium-chain lipid conjugation facilitates cell-permeability and bioactivity.* **J. Am. Chem. Soc.**, 2022, 144 (40), 18532-18544. [Link]
   Highlighted in commentaries by Nature and Science

SELECTED PROJECTS

Surface-Fingerprint-Based Geometric Deep Learning for Protein–Membrane Binding

- Developed MaSIF-PMP, a surface-based geometric deep learning framework enabling scalable prediction of membrane-binding interfaces without full MD simulations.
- Improved binding-site prediction accuracy by 30% compared to prior state-of-the-art methods and achieved MD-level accuracy in identifying interfaces.
- Engineered a validation pipeline leveraging accelerated MD simulations to verify model predictions and generate robust IBS labels, achieving a 3× reduction in computational cost.

De Novo Design of Selective Binders for Peripheral Membrane Proteins

- Developed a multi-stage computational framework integrating geometric deep learning (MaSIF-PMP) with generative diffusion models (RFdiffusion) to design *de novo* protein binders targeting specific IBSs.
- Engineered a post-hoc filtering strategy using MaSIF-PMP scores and geometric descriptors (globularity) to identify candidates with high target affinity and minimal membrane interference, validated through enhanced sampling MD across three case studies.
- Established MaSIF-PMP and geometric priors as effective reward signals for future fine-tuning of generative models.

Accelerating Computational Design of Nanoparticle-Protein Interactions

- Collaborating with the groups of Victor M. Zavala and Jessi Cisewski-Kehe.
- Designed geometric-invariant Topological Data Analysis (TDA) representations to robustly characterize complex protein structural features, ensuring model stability across varying coordinate systems.
- Achieved comparable prediction accuracy for protein–protein interaction interfaces while reducing data preprocessing time by 7× and training time by 6×.
- Extending the pipeline to analyze time-series dynamics, currently utilizing the model to characterize nanoparticle-protein interaction variance within MD simulations.

Context-Dependent Hydrophobic Interactions of Amphiphilic α-Peptides

- Collaborative work with Nicholas L. Abbott Group.
- Seeking to understand how hydrophobic interactions of α-peptides vary depending on sequence features and chemical context.
- Performed enhanced sampling methods (*e.g.,* steered molecular dynamics, replica exchange, umbrella sampling, metadynamics) to characterize mechanistic and energetic difference in interaction between coiled peptides and nonpolar surface.

PRESENTATIONS

Oral Presentations

- **B. Park**, R. C. Van Lehn. *Integrating Geometric Deep Learning and High-Throughput Molecular Simulations to Predict Protein-Membrane Binding Interfaces.* AIChE 2025, Boston, MA, USA (Nov 2025).
- **B. Park**, R. C. Van Lehn. *Integrating Geometric Deep Learning and High-Throughput Molecular Simulations to Predict Protein-Membrane Binding Interfaces.* Lab seminar invite at UW-Madison Cisewski Kehe Lab (Mar 2025).
- **B. Park**, R. C. Van Lehn. *Mechanistic and thermodynamic characterization of dynamic topology in an unassembled transmembrane protein.* AIChE 2024, San Diego, CA, USA (Oct 2024).
- **B. Park**, R. C. Van Lehn. *Mechanistic and thermodynamic characterization of dynamic topology in an unassembled transmembrane protein.* Lab seminar invite at UW-Madison Quentin Dudley Lab (Oct 2024).

Poster Presentations

- **B. Park**, C. Qiu, N. L. Abbott, R. C. Van Lehn. *Context-Dependent Hydrophobic Interactions of α-Peptides.* MTSM 2025, Madison, WI, USA (Jun 2025).
- **B. Park** and R. C. Van Lehn. *Surface-Centered Approach for Characterization and Prediction of Protein-Membrane Interactions.* AIChE 2024, San Diego, CA, USA (Oct 2024).
- **B. Park**, R. C. Van Lehn. *Mechanistic and thermodynamic characterization of dynamic topology in an unassembled transmembrane protein.* MTSM 2024, Ann Arbor, MI, USA (Jun 2024).
- **B. Park**, R. C. Van Lehn. *Mechanistic and thermodynamic characterization of dynamic topology in an unassembled transmembrane protein.* BPS 2024, Philadelphia, PA, USA, (Feb 2024).
- **B. Park** and R. C. Van Lehn. *Mechanistic and thermodynamic characterization of dynamic topology in an unassembled transmembrane protein.* MTSM 2023, Notre Dame, IN, USA (Jun 2023).

SKILLS

Programming: Python, Linux, Bash
ML Frameworks: PyTorch, TensorFlow, Scikit-learn, Graph Neural Networks, Transformers, Diffusion Models
MD Simulations: GROMACS, PLUMED, MDTraj, Umbrella Sampling, Replica Exchange, Metadynamics
High-Performance Computing: SLURM, MPI, GPU acceleration for ML and simulations
Tools: Docker, Git, LaTeX, AlphaFold, RFDiffusion, ProteinMPNN, PyMol, VMD, RDKit
Languages: Korean (native), English (fluent), Japanese (basic)

AWARDS

KFAS Overseas PhD Scholarship, Korea Foundation for Advanced Studies — Jul 2021 — Jun 2026
Full fellowship covering living expenses for outstanding academic and research performance (5 yrs., $65,000, <40 awardees)
Sinyang Scholarship, Sinyang Cultural Foundation — Mar 2019 — Dec 2020
Full tuition for academic excellence
Dean's List, Seoul National University — Feb 2019
Merit-based Scholarship, Seoul National University — Sep 2014 — Jul 2016

OTHER EXPERIENCES

Vice President & Treasury Officer, Korean Students & Scholars Association UW-Madison — Aug 2024 — Jul 2025
Career Development Officer, Korean Students & Scholars Association UW-Madison — Aug 2022 — Jul 2024
Coordinated visiting recruit events of corporates from South Korea (Samsung, LG, Hanwha, Lotte) for Korean graduate students and post-docs of UW-Madison.
Club Captain, Madison Korean Tennis Club — Mar 2024 — Mar 2026
Volunteer Staff, ISMB 2022 — Jul 2022
Demonstration Table Staff, UW-Madison 2022 Engineering Expo — Apr 2022
KIA (killed in action) Identification Agent, Korean MND Agency for KIA Recovery & Identification — 2016 — 2018